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Offline signature verification using a region based deep metric learning network

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ABSTRACT

Handwritten signature verification is a widely used biometric for person identity authentication in document forensics. Despite the tremendous efforts in past research, offline signature verification still remains a challenge, particularly in discriminating between genuine signatures and skilled forgeries, because the difference of appearance between genuine and skilled forgery may be smaller than that between genuine ones. This challenge is even more critical in writer-independent scenario, where each writer has very few samples for training. This paper proposes a region based Deep Convolutional Siamese Network using metric learning method, which is applicable to both writer-dependent (WD) and writer-independent (WI) scenario. For representing minute but discriminative details, a Mutual Signature DenseNet (MSDN) is designed to extract features and learn the similarity measure from local regions instead of whole signature images. Based on local regions comparison, the similarity scores of multiple regions are fused for final decision of verification. In experiments on public datasets CEDAR and GPDS, the proposed method achieved state-of-the-art performance of 6.74% EER and 8.24% EER in WI scenario, respectively, and 1.67% EER and 1.65% EER in WD scenario, respectively.

1. Introduction

As one of biometric traits, handwritten signature has been used to verify the identity of person in many applications for enhanced security and privacy. Compared with other measures of biometric authentication, signature verification is non-invasive and common in daily life, and therefore, has huge potential applications in legal, financial and administrative areas [1]. So far, a lot of research works have been reported for signature verification [2,3].

Depending on the device of signature data acquisition, signature verification techniques can be categorized into two types: online and offline. For online signature verification, dynamic information of signing process, such as pen trajectory, inclination and pressure, are usually utilized in feature extraction for verification [4–6]. In offline signature verification, signature image is captured by scanner or camera after writing is completed, and images are analyzed for verification [7–9]. Due to the missing of dynamic information of stokes, offline signature verification is more difficult than online verification. However, due to the popularity of paper documents, offline signature verification is more useful and widely needed in real applications.

The methods of signature verification can also be dichotomized into writer-dependent (WD) and writer-independent (WI) ones. In WD methods, a specialized metric model is learned for each individual writer during training stage, and then the learned metric model is used to classify the signature as genuine or not [10–12]. Obviously, WD systems need to be trained for every user, and relies on appreciable number of samples for each user. In real applications, it is unrealistic to collect a large number of samples per user. In WI methods, there is only one metric model for all writers, and the model trained with one set of writers can be generalized to disjoint new set of writers [9,13,14]. This is more realistic in practical applications, and using the samples of many writers for training a model can alleviate the over-fitting problem.

According to the degree of simulation, signature forgeries are categorized into random forgery, simple forgery and skilled forgery. A random forgery is signed without any information about the imitated writer. A simple forgery is the copy of the signature of the imitated person without following the writing style, while a skilled forgery is signed with effort to imitate the writing style of the genuine signature [1]. With high similarity of writing style to genuine signature, skilled forgery is very difficult to be verified. Ran-

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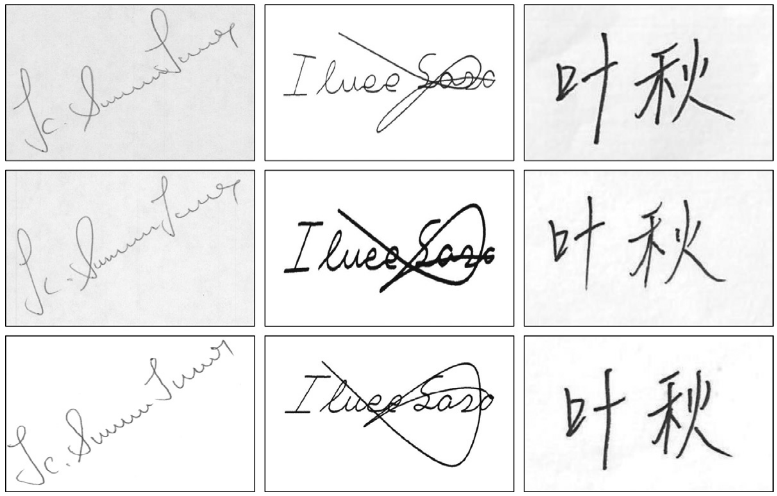


Fig. 1. Some genuine signature samples (first two rows) and skilled forgeries (third row).

dom forgeries are easily recognizable while they may often occur in commercial transactions, bank procedures [15,16].

Fig. 1 shows some signature samples. Genuine samples are shown in the first two rows while skilled forgeries are in the third row.

The difficulties of signature verification are mainly due to high intra-personal variability, limited number of training samples, scarcity of skilled forgery samples. Moreover, the difference between skilled forgery and genuine signature is subtle since forgers attempt to mimic genuine signature. With small inter-class variation between genuine signature and its (skilled) forgeries while larger intra-class variation of genuine signatures from one person, it is very hard to differentiate between intra-person signature genuineness and inter-personal forgery.

So far there have been many methods proposed for off-line signature verification, most of them concentrate on feature extraction/representation and similarity/distance metric evaluation. For feature extraction, various descriptors based on graphology, computer vision, signal processing, etc., have been proposed, such as local granulo-metric size distributions [17], directional properties of signature contours [18], energy of Curvelet coefficient [7], surroundedness features [9], Bag-of-Visual-Words (BoVW), Fisher vector and vector of locally aggregated descriptors (VLAD) [12]. Based on feature extraction, some works use a trained classifier to classify different writers or discriminate genuine against forgery. Among the popularly used classifiers is the support vector machine (SVM) [7,9,10,12,19,20]. Other classifiers used are neural network [9], Hidden Markov Model [21], and ensemble of classifiers [8].

Recent works take advantage of the feature learning ability of deep neural networks, particularly the convolutional neural network (CNN) [10,11,22]. The convolutional Siamese network, inputting two signature images, can learn visual features jointly with the similarity measure [14,23]. By learning features, these works have reported superior performance. However, they usually use high level features only, and so, cannot discriminate well skilled forgeries, which usually differ from genuine signatures in local characteristics, such as ends and bending of strokes. Therefore, considering features in local regions can be beneficial to signature verification.

To explore local features in offline signature verification, we propose a framework using a region based Deep Convolutional Siamese Network for metric learning. For feature extraction from images, we design a Mutual Signature DenseNet (MSDN), which combines high and low level features so as to obtain discriminative features for verification. Besides, local regions of signature image are input to the MSDN for extracting detailed features. The similarity measures learned from multiple local regions of whole signature image are fused to make final decision of genuine/forged signature. By training the network on signature samples of a set of writers, verification can be done on disjoint new writers, in so-called writer-independent (WI) scenario. It can also be applied to writer-dependent (WD) verification, by simply tuning the thresh for each new writer, unlike the WI method uses a global threshold. In experiments on two public datasets CEDAR and GPDS, our proposed method achieved 6.74% EER and 8.24% EER in WI scenario, respectively, and 1.67% EER and 1.65% EER in WD scenario, respectively. These results are competitive to the ones reported in the literature, and demonstrate the effectiveness of the proposed method. We also report results on a large dataset of Chinese signatures, named ChnSig.

A preliminary paper of this work, with results of only WI scenario, was presented at the conference [24]. Compared to the network used in Liu et al. [24], this paper designs a network structure called Mutual Signature DenseNet (MSDN) for enhanced feature extraction. The MSDN uses squeeze-and-excitation (SE) blocks [25] to interact features between two input signatures, so as to improve the verification performance. Besides, experimental results on WD scenario are added.

The rest of this paper is organized as follows: Section 2 reviews related work. Section 3 describes in details the proposed method. Section 4 presents experimental results, and Section 5 offers concluding remarks.

2. Related work

Signature verification can be considered as a two-class classification problem: to judge whether two signatures are signed by the same person or not, or to classify whether a signature is genuine for a specific user or not. In the past decades, many methods have been proposed for signature verification [2,3]. Previous non-deep-learning based methods usually involve steps of preprocessing, feature extraction and similarity measuring. The independent design of feature extraction and similarity/distance metric can not guarantee the optimality of verification performance. Nevertheless, the proposed preprocessing and feature extraction techniques are useful.

For signature feature extraction, various descriptors based on graphology, computer vision, signal processing, etc., have been proposed. For examples, Sabourin et al. [17] proposed local granulometric size distributions for representing signatures. Directional properties of signature contours are encoded while the length of regions enclosing letter is considered [18]. Energy of curvelet coefficient computed from signature image are used as feature representation [7]. The surroundedness features [9] characterize both shape and texture property of signature image. Okawa [12] proposed to represent signatures as Bag-of-Visual-Words (BoVW), Fisher vector and vector of locally aggregated descriptors (VLAD) based on local KAZE feature extraction on both foreground and background.

As for classification or comparing signature pairs, some previous methods used simple distance/similarity measures such as Euclidean distance or cosine similarity on feature vectors. In some works, a classifier was trained to classify different writers or discriminate genuine against forgery. Among the popularly used classifiers is the support vector machine (SVM) [7,9,10,12,19,20]. Other types of classifiers used are neural network [9], Hidden Markov Model [21], and ensemble of classifiers [8]. Methods of structural matching [26] and combining structural matching and neural network [27] also have been used in signature verification. A novel point-to-set (P2S) metric was proposed for offline signature verification, which divides a training batch into a support set and query set to pull each query set to its belonging support set [28].

Recent works of signature verification mostly use deep neural networks, particularly, the convolutional neural network (CNN) for learning features. Deep neural networks allow automatic feature extraction by hierarchical layer computing. The CNN was specially designed for learning features from images, by simulating local receptive fields in biological vision systems. CNNs have yielded superior performance in various vision tasks including image classification, object detection and segmentation. In previous works of signature verification, some methods first learn visual features using CNN, and then train a separate classifier (SVM or other type) for decision making, such as the methods [10,11,22]. Hafemann et al. [10] add spatial pyramid pooling to CNN to get fixed-sized feature representation from images of various sizes. Zheng et al. [22] extract displacement features from CNN to be combined with pooled convolutional features.

The Siamese neural network was first proposed in Bromley et al. [4] for online signature verification. This network has two branches of identical structure and shared parameters for feature extraction for two signatures, and the distance metric is trained jointly with the feature extractor. In recent works, the convolutional Siamese network, with its more powerful visual feature learning ability, is used to learn features and similarity metric for signature verification [14,15,23]. The two-channel CNN [29], inputting two signature images into one network as two channels, yields competitive performance to the convolutional Siamese network. To enhance the discrimination ability of features, Zheng et al. [22] proposed to extract displacement features.

As deep neural networks usually relies on a large number of training samples for good generalization, in signature verification, the CNN is usually used in writer-dependent (WD) verification where each user has many genuine and forgery samples, or the samples of many users are pooled to train a network (as in Siamese network) for writer-independent (WI) verification. Signature synthesis has been considered for augmenting training data for CNN [15], which applies Siamese network framework and focuses on random forgery verification.

The proposed method in this paper uses a Siamese network with DenseNet for feature extraction and combines the similarity scores of local regions for verification. The network structure and the region based score fusion are shown to be effective for improving the verification performance.

3. Proposed method

The diagram of the proposed method is shown in Fig. 2. The verification system consists of steps of preprocessing, region segmentation, feature extraction and metric model. The pair of signature images (to be verified whether they are signed by the same user or not) first undergo preprocessing, and each image is segmented into a series of overlapping regions. Then, the local region images are fed into the convolutional feature extractor network to obtain feature representations. The feature vectors of two signatures are fed into the metric model for calculating the similarity scores of corresponding local regions. The scores of local regions are fused for final decision. In training, the parameters of feature extractor and metric model are learned jointly with the objective of high intra-person similarity and low inter-person (between genuine and forgery) similarity.

3.1. Preprocessing

Preprocessing of signature image is to alleviate the influence of background, grayscale variation, size and location of signature on the verification performance. Our preprocessing procedure is shown in Fig. 3, consisting of four steps: tilt correction, background removal, foreground grayscale normalization, and size normalization (using moment normalization).

The input color image is firstly converted into grayscale one. Since many signature image samples are rotated, tilt correction is performed using the method proposed in Kalera et al. [30]. Otsu's binarization method [31] is applied to obtain the foreground mask, which is used to remove signature image background (label background pixels as 255). This eliminates the influence of the variation of background intensity values. For overcoming the influence of variation of foreground intensity, grayscale normalization is employed to standardize the mean and variance of foreground pixel values:

$$g'_f = \frac{(g_f - E(g_f)) \cdot v}{\delta(g_f)} + u, \tag{1}$$

where g_f and g'_f denote the gray level of original and normalized foreground pixel, respectively, $E(g_f)$ and $\sigma(g_f)$ denote the mean and standard deviation of gray level of foreground pixel. u and v are constants corresponding to the mean and standard deviation of normalized gray levels, which are empirically set as 30 and 10, respectively, in our experiments.

After background removal and grayscale normalization, size normalization is performed so as to standardize the size and location of signature in an image of fixed size. We adopt the moment normalization method [32] to fulfil this task since it is stable against the background noise and outlier strokes. The method is elaborated below.

Let $f(x, y)$ and $f'(x', y')$ denote the signature image (background eliminated) and the one after size normalization, respectively. By normalization, the pixel value $f(x, y)$ is mapped to $f'(x', y')$ by coordinate transformation:

$$x = \frac{x' - x'_c}{\alpha} + x_c, \tag{2}$$

$$y = \frac{y' - y'_c}{\alpha} + y_c, \tag{3}$$

where (x'_c, y'_c) denotes the center of normalized image, (x_c, y_c) denotes the centroid of foreground pixels of input image, and α is the ratio of scaling that can be estimated from moments:

$$\alpha = r \cdot \min\left(\frac{H_{norm}\sqrt{\mu_{00}}}{2\sqrt{2\mu_{02}}}, \frac{W_{norm}\sqrt{\mu_{00}}}{2\sqrt{2\mu_{20}}}\right), \tag{4}$$

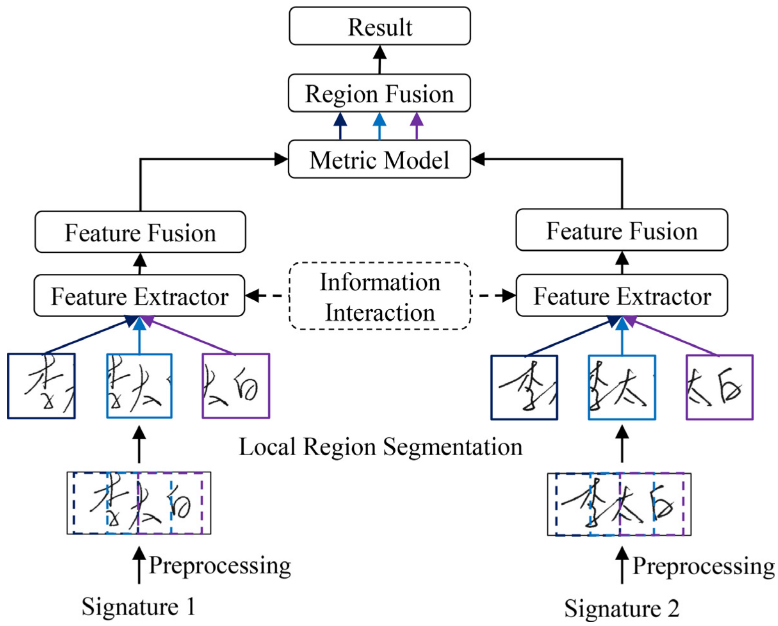


Fig. 2. The framework of the proposed region based metric network.

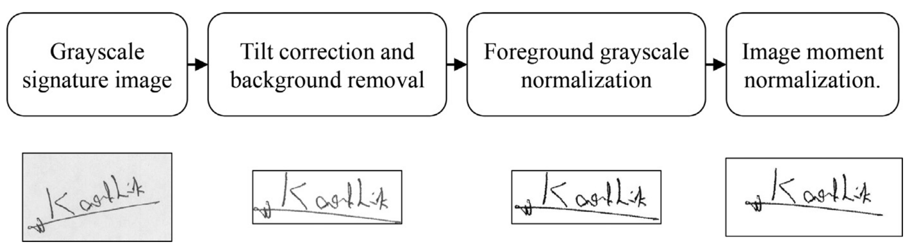


Fig. 3. Preprocessing steps with a sample image.

where r is the ratio of signature size to canvas size, H_{norm} and W_{norm} denote the pre-specified height and width of normalized image, and μ_{pq} denotes the central moments:

$$\mu_{pq} = \sum_x \sum_y (x - x_c)^p (y - y_c)^q [255 - f(x, y)]. \tag{5}$$

H_{norm} and W_{norm} are empirically set as 224 and 512, which may guarantee that the normalized signature image is of sufficient resolution. The ratio r is empirically set as 0.6 such that the normalized signature is mostly inside the canvas of normalized image. The coordinates (x, y) resulted from Eq. (2)(3) may be continuous, so the pixel values $f'(x', y')$ are mapped from $f(x, y)$ by bilinear interpolation.

Fig. 4 shows some examples of preprocessing procedure. After preprocessing procedure, gray levels of foreground pixels, position and size of signature are normalized, which are very important for verification performance.

3.2. Feature extraction

Feature representation is critical to the verification performance. For off-line signature verification, most existing methods extract features from the whole signature image [10,22]. However, the distinction of writing style between genuine signature and skilled forgery usually lies in the details contained in stroke parts or local regions. Such details can be better grasped by Siamese networks on local regions than a network on the whole image. So, we propose to learn features from local regions and then fuse the similarity measures of multiple local regions.

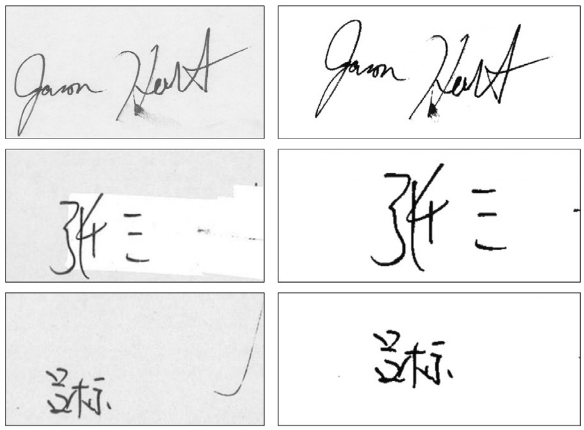
In our previous work [24], the input signature image is firstly segmented into several local regions, from which features are extracted by CNNs and then fused. It was justified that the fusion of local features performs better than extracting features from the whole signature image. Fig. 5 shows the examples of local region segmentation. From the normalized signature image of size 512×224, we empirically extract nine regions of size 224×224 from left to right with step 36 and horizontal overlap of 188 pixels.

In order to further improve the verification performance, in this paper we design a Mutual Signature DenseNet (MSDN), which unifies high-level and low-level features and uses the local region as the input so as to obtain more discriminative features.

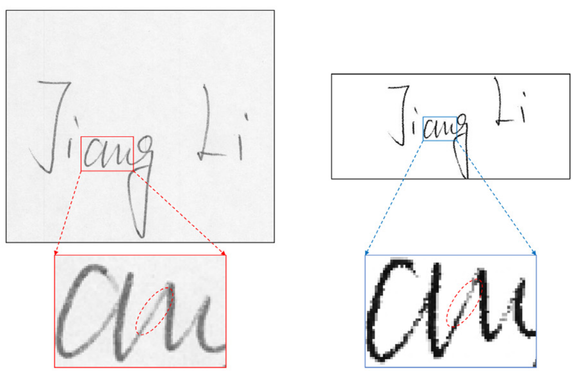
The proposed MSDN consists of two CNN branches sharing the same structure and parameters to learn the feature representation of signature. SE-Blocks (squeeze-and-excitation blocks) [25] and Spatial Pyramid Pooling (SPP) are combined into two CNN branches so as to interact information of two inputs. In this way, the characteristics of two signature images can be fully utilized.

Popular CNN architectures, such as AlexNet [33], VGG [34], ResNet [35] and DenseNet [36], have been widely use in various computer vision tasks. We compared some representation architectures (Section 4.2), and finally choose the DenseNet as the backbone to constitute the mutual network.

The structure of MSDN is shown in Fig. 6. Specifically, SE-Blocks (squeeze-and-excitation blocks) [25] are added into each DenseBlock to interact information of two inputs. This makes the learned weights better account for the difference between two input signatures. The outputs of each DenseBlock are first converted into a



(a)



(b)

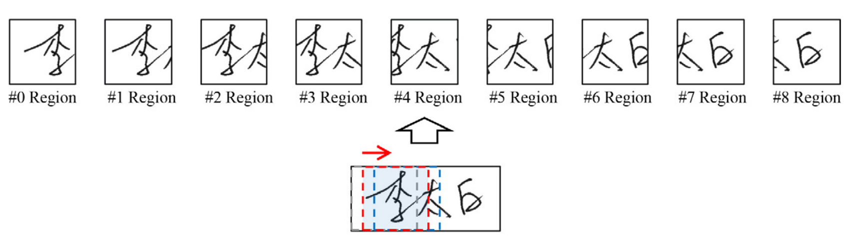
Fig. 4. Examples of preprocessing procedure. Original signature images are in the left and the preprocessed images are in the right.

Table 1
The configuration details of MSDN.

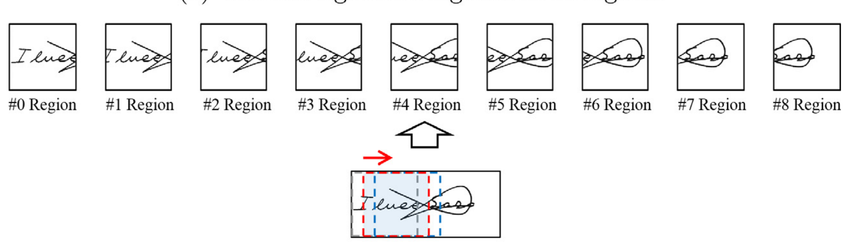
Layers	Parameters
Convolution	7×7 conv, channel 64, stride 2
Max Pooling	3×3 max pool, stride 2
DenseBlock(1)	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 3$
SE Block(1)	$160 \rightarrow 40 \rightarrow 160$
Transition(1)	1×1 conv, channel 80
	2×2 average pool, stride 2
DenseBlock(2)	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 4$
SE Block(2)	$208 \rightarrow 52 \rightarrow 208$
Transition(2)	1×1 conv, channel 104
	2×2 average pool, stride 2
DenseBlock(3)	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$
SE Block(3)	$296 \rightarrow 74 \rightarrow 296$
Transition(3)	1×1 conv, channel 148
	2×2 average pool, stride 2
DenseBlock(4)	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 3$
SE Block(4)	$244 \rightarrow 61 \rightarrow 244$
Convolution 1x1	1×1 conv, channel 512
SPP	output sizes $\begin{bmatrix} 1 \times 1 \\ 2 \times 2 \\ 4 \times 4 \end{bmatrix}$

vector by global average pooling. Then, the weights for the channels of DenseBlock's output are calculated from the difference between the two signature images by fully connected layers with sigmoid activation function. Finally, the features extracted by different transition layers are resized by average pooling into the size of the last DenseBlock's output, to be combined with a convolution layer and then converted into the final signature feature representations using Spatial Pyramid Pooling (SPP). The configuration details of MSDN are given in Table 1. We do not set dropout but add batch normalization for each convolution layer. The number of channels of the first convolution layer is set as $N_{init} = 64$, and growth rate set as $k = 32$ following the way in Huang et al. [36].

The input of MSDN can be either the whole signature image or a local region, since SPP enables feature extraction from images of different sizes. When the input image is the whole signature image (size 512×224), each channel before SPP is of 16×7; while inputting a local region (size 224×224), each channel is of size 7×7. From the feature map of either 16×7 or 7×7, SPP is performed to extract three feature maps of size 1×1, 2×2 and



(a) Chinese signature segmented in regions



(b) Western signature segmented in regions

Fig. 5. Region segmentation of signature image.

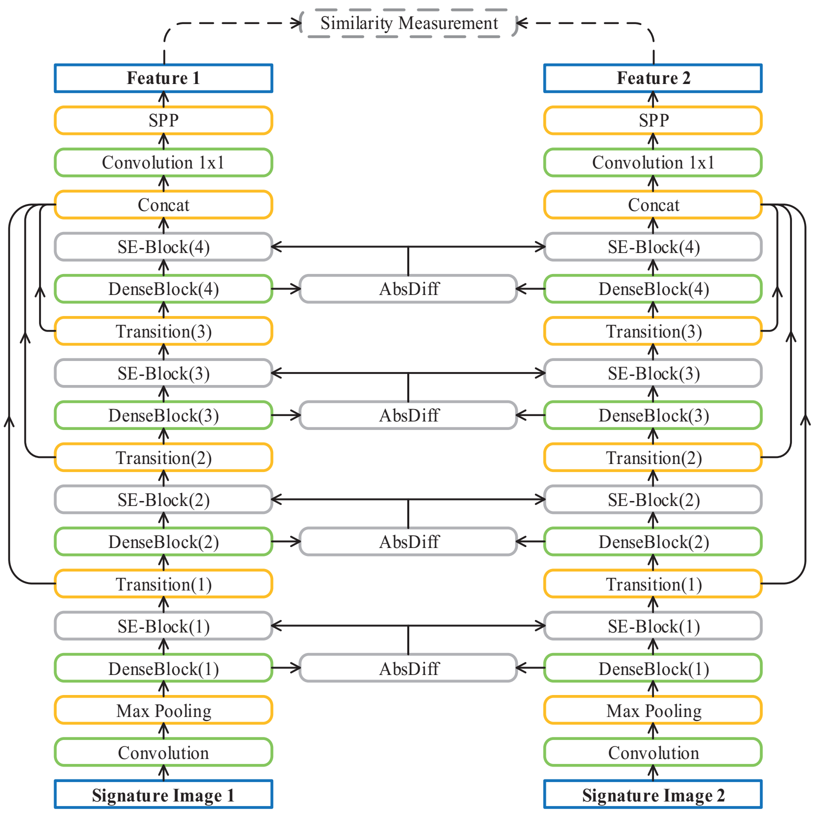


Fig. 6. The structure of MSDN.

4×4. And so, the output feature vector dimensionality for an input whole/region image is $512 \times (1 + 4 + 16) = 10,752$.

For training the model for region-based verification, different local regions are used as samples to be processed by the same network so that only one network is trained for feature extraction from different regions.

3.3. Region based metric learning

In the proposed method, the between-signature similarity is obtained by fusing the similarity measures of multiple local regions, while the similarity metric for a local region (between corresponding regions of two signatures) is learned using the Siamese network [4,24] with feature extraction using the MSDN.

A pair of feature vectors output by MSDN are denoted as S and T, respectively, where $S, T \in \Re^d$, d is the dimensionality of feature vector. The difference between the pair feature vectors S and T, denoted as F, reflects the similarity of the two inputs. We have tried to compute F using different ways, such as cosine similarity, Euclidean distance, and absolute difference. The absolute difference vector is found to perform best.

On the absolute difference vector F, a fully connected layer network with sigmoidal non-linearity is applied to calculate the similarity measure, which represents the predicted probability that the two input signatures (or corresponding local regions) belong to the same writer:

$$\hat{p} = sigmoid(W \cdot F + b), \tag{6}$$

where $\hat{p}$, W and b denotes the predicted probability, network weights and bias, respectively.

For network training (the feature extraction network and similarity metric model are trained jointly), the loss function is the cross-entropy loss:

$$Loss(t, \hat{p}) = -\Big[t \cdot ln(\hat{p}) + (1 - t) \cdot ln(1 - \hat{p})\Big], \tag{7}$$

where t is the target value. $t = 1$ if the two input signature images are from the same writer, otherwise $t = 0$. The loss is summed over all local regions of signature samples.

3.4. Signature verification decision

In verification (testing) stage, each pair of local regions (from two input signatures) contributes a probability based on the feature difference vector. The probabilities of different local regions are fused by averaging to make the final decision:

$$P = \frac{1}{n} \sum \hat{p}_i, \tag{8}$$

where n is the number of local regions in a signature, $\hat{p}_i$ denotes the probability of each local region. Finally, the decision of signature verification is decided by

$$decision = \begin{cases} \text{accept if } P > thr \\ \text{reject if } P \leq thr \end{cases} \tag{9}$$

where thr is threshold, which is adjustable to balance False Accept Rate (FAR) and False Reject Rate (FRR).

4. Experiments

4.1. Experimental setup

Two public offline handwritten signature datasets, CEDAR [30] and GPDS Synthetic [37] are used in our experiments. In order to evaluate the verification performance on Chinese signature images as well as for potential application in Chinese scenario, we established a Chinese signature image dataset named ChnSig. Therefore, we conducted experiments on three datasets, CEDAR, GPDS Synthetic and ChnSig. Skilled forgeries are available

Table 2
Details of experimental datasets.

Datasets	#users	Training #samples	tTest #samples	Language
CEDAR	55	42,600	4260	English
GPDS	4000	1,992,000	1,992,000	English
ChnSig	1243	205,000	49,815	Chinese

in all the three datasets. Table 2 gives the details of the three datasets.

Specifically, the CEDAR dataset contains offline signature images of 55 users. The users were asked to create forgeries of signatures from other writers. Each user has 24 genuine signatures and 24 skilled forgeries. So, we can get $C_{24}^{2}=276$ genuine-genuine pairs of signatures as positive samples and $C_{24}^{1}\times C_{24}^{1}=576$ genuine-forged pairs of signatures as negative samples for each user. We randomly selected 50 user as training set and the remaining 5 users as test set. Totally, there are 42,600 samples for training and 4260 samples for testing.

The GPDS Synthetic dataset was created with samples for 4,000 synthetic signers. The signatures were synthesized to approximate the characteristics of real signatures closely, and the verification performance on the synthetic dataset was also similar to that on real dataset [37]. In the GPDS Synthetic dataset, each user has 24 genuine signatures and 30 skilled forgeries. So, we can get $C_{24}^{2}=276$ positive samples and $C_{24}^{1}\times C_{30}^{1}=720$ negative samples per user. We randomly selected 2000 users as training set and the remaining 2000 users as test set. Totally, we get 1,992,000 samples for training and 1,992,000 samples for testing.

The ChnSig dataset is a Chinese offline signature dataset created by ourself from 1,243 users. Each user has 10 genuine signatures and 16 skilled forgeries. 1000 users are randomly selected to form training set while the remaining 243 users are used to form the test set. Totally, 205,000 samples are collected for training and 49,815 samples for testing.

Since deep neural networks usually need large number of samples for training while both CEDAR and ChnSig dataset have relatively small number of samples, for these two datasets, we use the model pre-trained with GPDS dataset and then fine-tune on the CEDAR/ChgSig dataset.

Our model was implemented on the platform of PyTorch [38]. The training algorithm was implemented using Adam [39] with learning rate 0.001. We used mini-batches of 64 pairs of signature regions (32 for whole images) in training. For the metric model, dropout was set as 0.3, but the feature extractor did not set dropout in training. Experiments were performed on a workstation with Intel(R) Xeon(R) E5-2680 CPU, 256 GB RAM and a NVIDIA GeForce GTX TITAN X GPU. The system takes 258 ms to verify a pair of signatures on average.

We report the False Reject Rate (FRR) of genuine (misclassifying a genuine as forgery), the False Accept Rate (FAR) of forgeries (misclassifying a forgery as genuine) and the Equal Error Rate (EER) in experiments. EER is the error rate when FRR is equal to FAR, which can be obtained by adjusting the decision threshold. For calculating EER, we consider two ways of selecting decision threshold: for writer-dependent (WD) verification, the threshold is adjusted for each user for achieving equal error rate on the specific user; for writer-independent (WI) verification, the threshold is shared for all users and adjusted to achieve overall equal error rate on all users. Since the verification of skilled forgeries is most difficult and some datasets do not provide samples of random and simple forgeries, we report results of discriminating skilled forgeries only, following most previous works. In the following sections, we report results of WI verification when comparing different configurations of the proposed model, and then report results of WD verification for comparison with state of the art.

Table 3
Comparison of CNN architectures on GPDS dataset (WI verification on whole signature image.

Architecture	EER (%)	Model size
AlexNet	14.13	9.9 MB
ResNet-18	11.74	11.4 MB
VGG-16	10.95	14.8 MB
DenseNet-36	**10.93**	**4.2** MB

4.2. Comparison of CNN architectures

In the signature verification model (Fig. 2), the feature extraction network is a CNN, which can be different architectures. In the scenario of writer-independent (WI) verification by feature extraction from whole signature image (512×224), we compare the performance of popular architectures AlexNet, ResNet-18, VGG-16, and DenseNet-36. In this experiment, the network without SE-Blocks (same as the one in Liu et al. [24]) is used, so as to evaluate the performance of CNN architectures. The experimental results are given in Table 3. It can be seen that the DenseNet-36 achieves the best performance. Besides, compared with VGG-16, ResNet-18 and AlexNet, DenseNet-36 has the smallest model size. Consequently, the proposed MSDN is constructed by using DenseNet-36 as backbone.

4.3. Effectiveness of region fusion scheme

For feature extraction from local regions, we extract nine regions of 224×224 from the normalized signature image (size 512×224) with step 36 and overlap of 188 pixels, as shown in Fig. 5. The local regions are numbered as $\{0,1\ldots,8\}$. The signature verification model (including feature extraction network and metric model) is trained using the nine regions of training signatures as samples, and applied to verification by fusing the similarity measures of various combinations of local regions.

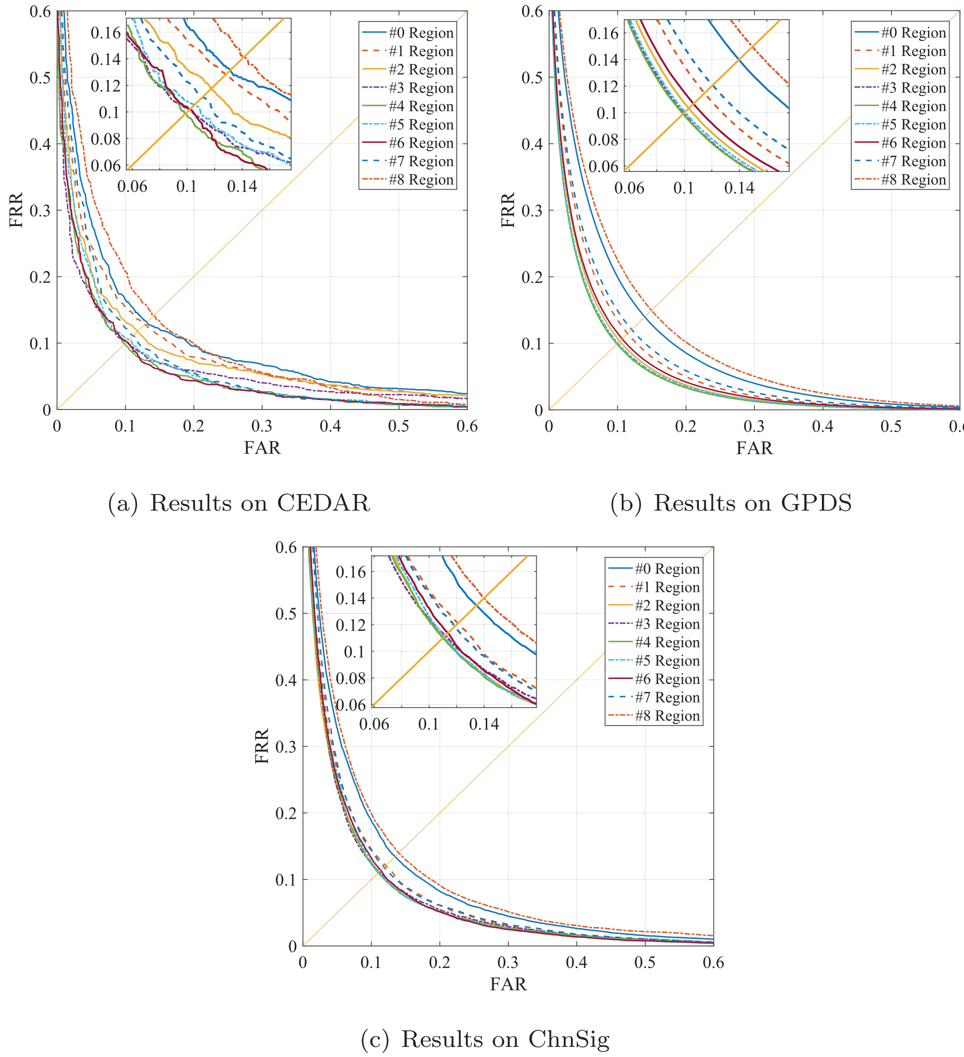
First, we evaluate the verification performance using individual local region. The results of different individual regions are shown in 4, and the ROC curves are shown in Fig. 7. From the results, we can see that the best performance is given by the central region $\{4\}$. Besides, the local region the closer to the center of signature yields the better performance. This can be explained that the central part of signature image is most informative.

Based on the performances of individual local regions, next we compare the performance of difference combinations of local regions. We combine the central region with some neighboring regions on the left and right, and form five combinations: (1) all nine regions; (2) the central seven regions $\{1,2,3,4,5,6,7\}$; (3) the central five regions $\{2,3,4,5,6\}$; (4) central plus left/right most regions $\{0,4,8\}$; (5) central plus two $\{1,4,7\}$. Note that the central region 4 is overlapping with the left/right most regions 0 and 8, because the sampling step of regions is 36 while the region width 224 is greater than 4 times step. So, the combination $\{0,4,8\}$ covers the whole signature image. The results (EER) of these combinations are given in Table 5, and the ROC curves are shown in Fig. 8. It can be seen that all the combinations of local regions outperform the whole image based verification in accuracy, but processing multiple local regions may consume more computing time.

Among different combinations, the combination of all nine regions gives best EER on two of three datasets, and its EER on CEDAR is nearly the best. The combination $\{1,2,3,4,5,6,7\}$ performs slightly worse, indicating that the left/right most regions

Table 4
Signature verification performance using single regions (EER in %). $\{i\}$ denotes the order number of region from left to right.

Dataset	{0}	{1}	{2}	{3}	{4}	{5}	{6}	{7}	{8}
CEDAR	13.97	12.81	11.77	10.21	**9.86**	10.59	10.17	11.11	14.95
GPDS	13.19	11.47	10.32	9.94	**9.91**	10.04	10.67	12.06	14.13
ChnSig	13.43	12.08	11.04	11.17	**10.97**	11.09	11.37	11.84	13.99



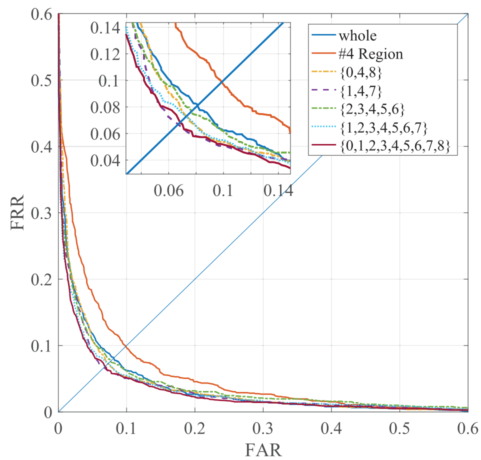
(a) Results on CEDAR
(b) Results on GPDS
(c) Results on ChnSig

Fig. 7. Verification results (ROC curves) of different single regions on three datasets.

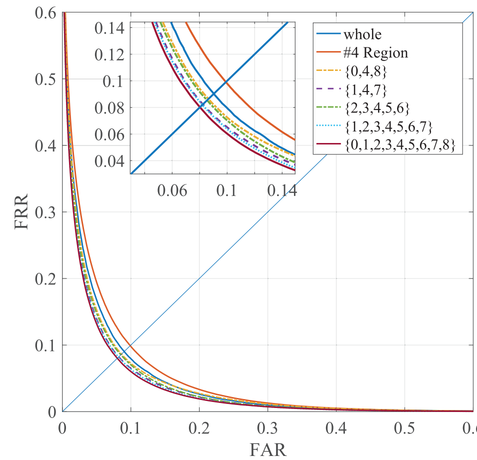
Table 5
Signature verification performance of different combinations of local regions (EER in %).

Dataset	Whole	All-nine	{1,2,3,4,5,6,7}	{2,3,4,5,6}	{0,4,8}	{1,4,7}
CEDAR	8.26	6.84	7.33	7.64	7.26	**6.74**
GPDS	8.99	**8.02**	8.20	8.54	8.64	8.24
ChnSig	10.21	**9.01**	9.20	9.54	9.50	9.22
time/pair	86 ms	142 ms	116 ms	102 ms	88 ms	88 ms

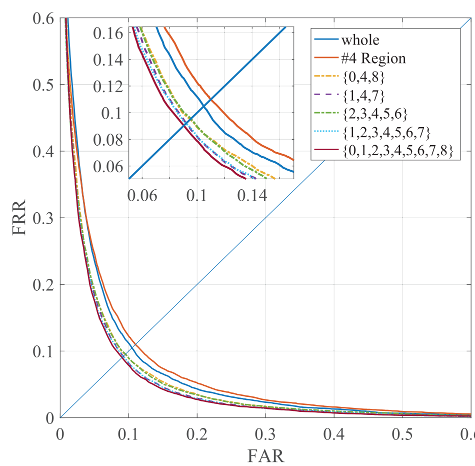
contain little information (actually there are few foreground pixels). The combination $\{1, 4, 7\}$ performs surprisingly well. Especially, it performs best on the dataset CEDAR and comparably with combining all nine regions on datasets GPDS and ChnSig. This indicates that regions 1 and 7 are sufficiently complementary to the central region 4. The combination $\{1, 4, 7\}$ also consumes little extra computing time compared to the whole image based verification. The combination $\{1, 4, 7\}$ performs comparably or better than the combination of more regions $\{1, 2, 3, 4, 5, 6, 7\}$ can be explained that the more regions contain redundant information while fusion by averaging (Eq. (8)) does not handle this redun-



(a) Verification results in CEDAR



(b) Verification results in GPDS



(c) Verification results in ChnSig

Fig. 8. Verification results (ROC curves) of different combinations of local regions.

Table 6
Signature verification performance of DenseNet combined with SE-Block and multi-level feature (MF) (EER in %).

Architecture	CEDAR	GPDS	ChnSig
DenseNet	**4.55**	8.89	9.91
DenseNet + SE	7.43	8.51	9.68
DenseNet + MF	10.21	9.61	10.04
DenseNet + SE + MF	6.74	**8.24**	**9.22**

Table 7
Signature verification performance using writer-dependent (WD) threshold (EER in %). Ref-n means using n references per writer.

Dataset	Ref-1	Ref-4	Ref-8	Ref-10	Ref-12
CEDAR	4.83	2.17	2.33	1.75	1.67
GPDS	6.49	2.90	2.05	1.80	1.65

dancy optimally. The combination three regions $\{1, 4, 7\}$ is used in the following experiments.

4.4. Comparison of MSDN configurations

As described in Section 3.2, we propose a MSDN consisting of two CNN branches for feature extraction, which combines high and low level features to construct multi-level feature representation of input. Moreover, the SE-blocks and SPP are applied to inter-act information between two inputs. We conduct experiments to justify the effects of SE-Block and multi-level feature (MF) as well as the combination of SE-Block and MF. The results on the three datasets are given in Table 6. The results are obtained by combining the similarity measures of three local region $\{1, 4, 7\}$.

From the results, we can see that compared with the original DenseNet, adding SE-Blocks improves the verification performance on two datasets GDPS and ChnSig. SE-Block makes use of the information difference between the two input signatures in the middle layer of the network, so as to better learn discriminative features. Multi-level feature representation does not improve the performance, as concatenating multi-level features may import more feature details and may leading to over-fitting. Nevertheless, SE-Blocks can overcome this effect, and so, combing SE-Blocks and MF can yield even better performance than DenseNet + SE. On two datasets GPDS and ChnSig, the configuration SenseNet + SE + MF gives the best results.

It is worth noting here that on the CEDAR Database, DenseNet gives the best result, outperforming DenseNet + SE + MF. As mentioned before, the size (number of samples) of CEDAR dataset is much smaller than the other two. On a small data, more com-

Table 8
Comparisons with other methods on CEDAR dataset. WI: writer-independent; WD: writer-dependent.

Method	Type	#user	#ref	Feature	EER (%)
Kumar et al. [40]	WI	55	1	morphology	11.59
Kumar et al. [9]	WI	55	1	Surroundedness	8.33
Xing et al. [14]	WI	55	1	Siamese Network	8.50
Zhu et al. [28]	WI	55	5	P2S metric	9.29
Proposed	**WI**	**55**	**1**	**MSDN**	**6.74**
Chen et al. [41]	WD	55	16	Zernike Moments	16.40
Chen et al. [42]	WD	55	16	Graph Matching	7.90
Guerbai et al. [7]	WD	55	4	Curvelet transform	8.70
Guerbai et al. [7]	WD	55	8	Curvelet transform	7.83
Guerbai et al. [7]	WD	55	12	Curvelet transform	5.60
Hafemann et al. [11]	WD	55	12	SigNet-F	4.63
Hafemann et al. [10]	WD	55	10	SigNet-SPP-300dpi	3.60
Zhu et al. [28]	WI	55	5	P2S metric	5.22
Proposed	WD	55	10	MSDN	1.75
Proposed	**WD**	**55**	**12**	**MSDN**	**1.67**

Table 9
Comparisons with other methods on GPDS dataset. WI: writer-independent; WD: writer-dependent.

Method	Type	#user	#ref	Feature	EER (%)
Kumar et al. [9]	WI	300	1	Surroundedness	13.76
Hu et al. [8]	WI	300	1	LBP + GLCM + HOG	9.94
Soleimani et al. [13]	WI	4000	1	HOG	13.30
Xing et al. [14]	WI	4000	1	Siamese Network	10.37
Li et al. [29]	WI	4000	1	Two-channel CNN	9.95
Proposed	**WI**	**4000**	**1**	**MSDN**	**8.24**
Ferrer et al. [43]	WD	160	12	Geometrical Features	13.35
Vargas et al. [44]	WD	160	10	HPPPD	12.33
Hu et al. [8]	WD	300	10	LBP + GLCM + HOG	7.66
Guerbai et al. [7]	WD	300	4	Curvelet Transform	16.92
Guerbai et al. [7]	WD	300	8	Curvelet Transform	15.95
Guerbai et al. [7]	WD	300	12	Curvelet Transform	15.07
Hafemann et al. [11]	WD	300	12	SigNet-F	1.69
Hafemann et al. [10]	WD	300	10	SigNet-SPP-300dpi	3.15
Proposed	WD	4000	10	MSDN	1.80
Proposed	**WD**	**4000**	**12**	**MSDN**	**1.65**

Table 10
Experimental results of random forgery verification of proposed method.

DataBase	AUC (%)	EER (%)
CEDAR	99.68	2.67
GPDS	99.93	1.26
ChnSig	99.65	1.92

plicated network and higher feature dimensionality may result in worse performance because of over-fitting on the training dataset. Also, the accuracy on a small test dataset is less confident than that on a large dataset.

4.5. Results of writer-dependent verification

The above results are reported for writer-independent verification, where a global threshold is used for all writers to get the FAR and FRR. For comparison with previous methods in writer-dependent (WD) scenario, we also report results of WD verification on three datasets. When applied to WD signature verification, the proposed method only needs to tune writer-dependent threshold, without fine-tuning the CNN feature extractor and metric model. To do this, we randomly selected some genuine signatures as references for each writer in the test set. For each test writer, the rest signatures are used as test samples to be compared with the references. After feature extraction and similarity scoring, the test sample is judged to be genuine or not based on the average score with multiple references. The threshold is tuned for each writer to get the EER. For each writer, references are selected randomly for 10 times, and the average EER is reported. The results on CEDAR and GPDS datasets (which were frequently used in previous works) are shown in Table 7.

Compared to the corresponding results of writer-independent verification in Table 6, we can see that writer-dependent verification gives much lower error rates. When using one reference per writer, the EER of WD verification on two datasets is 4.83/6.49%, compared to 6.74/8.24% of WI verification. When increasing the number of references per writer, the error rate is decreased progressively. Using 12 references per writer, the error rates are very low compared to the results of WI verification.

4.6. Comparison with previous methods

To compare with the performance of previous methods in the literature, we collected some representative results on the CEDAR and GPDS datasets. The results are shown in Tables 8 and 9, respectively. The results are categorized into ones of writer-independent (WI) setting and ones of writer-dependent settings (WD). In WI scenario, the writers of training and test set are disjoint, and in testing, the test signature is compared with one reference for decision. In WD scenario, usually samples from all writers are used in training, and each writer has a model or discriminant function to judge whether a test signature is signed by the specific writer or not. In our method for WD scenario, we train the feature extraction network using the data of writers disjoint from the test writers, but in testing, each test writer is estimated a threshold based on multiple references, without writer-dependent training of model.

Note that some methods compared did not use the same data setting with us. Some used different versions of GPDS dataset or the data of different number of writers in GDPS Synthetic dataset. Some partitioned the training and test sets of GPDS dataset in different ways. In experiments on CEDAR dataset, most methods used the data of 50 writers for training and the data of 5 writers for testing and repeated random partitioning multiple times. We follow this way. For experiments on GPDS Synthetic dataset, we followed the way of [9] and [14] to use half of writers for training and half for testing, but the dataset of [9] contains 300 writers instead of 4000. The experiment of [29] used 2500 writers in training and 1500 writers in testing. In Tables 8 and 9, some methods give average error rate (average of FER and FAR) instead of EER. The average error rate is comparable with EER, anyway.

On the CEDAR dataset, some previous works (such as [23,29]) have reported 0% EER. Those works did not perform preprocessing of background removal (binarization) on the signature images.[1] The 0% error can be attributed to the difference of background between genuine and forgery signatures. For generality, image background should be removed or normalized to avoid the disturbance of illumination variation. So, such results are not included in our tables.

From the results in Tables 8 and 9, we can see that in WI scenario, the proposed method outperforms the previous methods on CEDAR and GPDS datasets. Particularly, our performance is favorable to those of [9,14,29] in similar data partitioning. In WD scenario, our methods also performs superiorly to the previous ones though our method only tunes the threshold on the references of test writers without fine-tuning of network model, unlike that most WD methods train the model directly on the data of test writers.

4.7. Experiments on random forgery verification

Although random forgery is inherently easier verified compared with skilled forgery, it is much more often appear in application scenarios, such as access control systems, commercial transactions, banking documents [15]. Therefore, experiments on random forgery verification are run for performance evaluation.

For collection of random forgery testing samples, genuine signatures of other users from databases are randomly selected as random forgeries for test users. Specifically, 24 genuine signatures of other users from CEDAR and GDPS, 10 genuine signatures of other users from ChnSig, are randomly chosen. Finally, there obtained 2880, 1,152,000 and 24,300 random forgery testing samples corresponding to database CEDAR, GDPS and ChnSig, respectively.

The experimental results are listed in Table 10. Although the proposed method is mainly designed for skilled forgery verification, it achieves good performance on random forgery verification also.

As for comparison with other methods, it should be noted that comparisons are not always fair due to different numbers of samples used in training and testing procedures, etc. For reference, we compare with the method proposed in Ruiz et al. [15] since it applies the same framework (Siamese Network) and focuses on random forgery verification. The network proposed in Ruiz et al. [15] has been trained with samples of Database GPDS while tested on CEDAR. It reports 99.09% AUC and 4.84% EER on CEDAR database of random forgery verification. It can be seen that the proposed method is competitive.

[1] In our experiments, we could also achieve 0% EER if processing the signature images of CEDAR dataset without background removal.

5. Conclusion

In this paper, we proposed a novel framework for offline signature verification with region based deep metric learning network. For visual feature extraction, we designed a Mutual Signature DenseNet (MSDN), which extracts features in different levels and combines them, by learning from pairs of signatures. The network inputs either whole signature image or local regions of signature. The similarity scores of multiple local regions are fused to make decision of verification. We evaluated the method on three datasets (two public English ones and a Chinese dataset) with variable configurations. It is verified that fusing local regions can improve the verification performance because of better utilization of local features. Compared to state-of-the-art methods in the literature, our method performs superiorly in similar data partitioning, for either writer-independent or writer-dependent scenario, though the method was primarily designed for writer-independent verification for application generality. In future, further improvements can be made by exploiting better local features and combining with structural matching methods.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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